

SECTION A

Question				Marking details	Marks Available					
					Total	AO1	AO2	AO3	Maths	Prac
1.				$\text{Pb}^{2+}(\text{aq}) + 2\text{I}^{-}(\text{aq}) \rightarrow \text{PbI}_2(\text{s})$	1	1				1
2.				enthalpy of solution = $(-464) + (-364) - (-905) = 77 \text{ kJ mol}^{-1}$ (1) enthalpy of solution is positive/endothermic so it is insoluble (1) alternative method enthalpy of hydration = -828 kJ mol^{-1} (1) enthalpy of lattice formation is more exothermic so it is insoluble (1)	2	1	1			
3.				metal ion is lithium as it gives a red flame test (1) anion is iodide as it gives purple fumes (of iodine) / rotten egg smell (of hydrogen sulfide) (1)	2		1			1
4.				Al is electron deficient / does not have a full outer shell (1) lone pair on Cl (1)	2	1				
5.				royal blue	1	1				1
6.				catalyst in a different physical state to the reactants	1	1				
7.				boron nitride doesn't conduct electricity but graphite does must name both substances or state that 'only' graphite conducts electricity	1	1				
					10	7	3	0	0	5